1.

X (m, c, w, h) $\longrightarrow$ [1] convolution $\longrightarrow$ [2] max pooling $\longrightarrow$ [3] ReLu

$W^{[1]}$: (f, c, f_w, f_h) $\quad$ $O^{[2]}$: $(f, C_w/p, C_h/p)$ $\quad$ $O^{[3]}$: (f, P_w, P_h)

$b^{[1]}$: $(f, 1)$ $\quad$ $=(f, P_w, P_h)$

$O^{[1]}$: $(f, w-f_w+1, h-f_h+1)$

$=(f, C_w, C_h)$

$\longrightarrow$ [4] linear $\longrightarrow$ [5] Softmax $\longrightarrow \hat{y}$

$W^{[4]}$: $(10, \text{flat})$ $\quad$ $O^{[5]}$: $(10, 1)$

$b^{[4]}$: $(10, 1)$

$O^{[4]}$: $(10, 1)$

<Cost>

$$\mathcal{L}(y, \hat{y}) = -\sum_{k=1}^{K} y_k \log \hat{y}_k, \quad \therefore \frac{\partial \mathcal{L}}{\partial \hat{y}} = \begin{bmatrix} \partial \mathcal{L}/\partial \hat{y}_1 \\ \vdots \\ \partial \mathcal{L}/\partial \hat{y}_K \end{bmatrix} = \begin{bmatrix} -y_1/\hat{y}_1 \\ \vdots \\ -y_K/\hat{y}_K \end{bmatrix} = \frac{\partial \mathcal{L}}{\partial O^{[5]}}$$

<Softmax>

$$O_k^{[5]} = \frac{\exp(O_k^{[4]})}{\sum_{l=1}^{K} \exp(O_l^{[4]})} \qquad \frac{\partial \mathcal{L}}{\partial O_k^{[4]}} = \sum_{p=1}^{K} \frac{\partial \mathcal{L}}{\partial O_p^{[5]}} \frac{\partial O_p^{[5]}}{\partial O_k^{[4]}} = \sum_{p=1}^{K} \frac{\partial \mathcal{L}}{\partial O_p^{[5]}} \hat{y}_p (\delta_{pk} - \hat{y}_k)$$

<linear>

$$O_k^{[4]} = \sum_{j}^{\text{flat}} W_{kj}^{[4]} O_j^{[3]} + b_k^{[4]}$$

$$\frac{\partial \mathcal{L}}{\partial W_{ij}^{[4]}} = \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial O_k^{[4]}} \frac{\partial O_k^{[4]}}{\partial W_{ij}^{[4]}} = \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial O_k^{[4]}} \delta_{ki} O_j^{[3]} = \frac{\partial \mathcal{L}}{\partial O_i^{[4]}} O_j^{[3]}$$

$$\frac{\partial \mathcal{L}}{\partial O_i^{[3]}} = \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial O_k^{[4]}} \frac{\partial O_k^{[4]}}{\partial O_i^{[3]}} = \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial O_k^{[4]}} W_{ki}^{[4]}$$

$$\frac{\partial \mathcal{L}}{\partial b_i^{[4]}} = \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial O_k^{[4]}} \frac{\partial O_k^{[4]}}{\partial b_i^{[4]}} = \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial O_k^{[4]}} \delta_{ki} = \frac{\partial \mathcal{L}}{\partial O_i^{[4]}}$$

<Relu>

$$O_{pqr}^{[3]} = \max(0, O_{pqr}^{[2]}) \qquad \therefore \frac{\partial \mathcal{L}}{\partial O_{ijk}^{[2]}} = \sum_{pqr} \frac{\partial \mathcal{L}}{\partial O_{pqr}^{[3]}} \frac{\partial O_{pqr}^{[3]}}{\partial O_{ijk}^{[2]}} = \sum_{pqr} \frac{\partial \mathcal{L}}{\partial O_{pqr}^{[3]}} \cdot 1\{(p,q,r)=(i,j,k)\} \cdot 1\{O_{ijk}^{[2]} > 0\}$$

$$= \frac{\partial \mathcal{L}}{\partial O_{ijk}^{[3]}} 1\{O_{ijk}^{[2]} > 0\}$$

< max pooling >

$$O^{[2]}_{pqr} = \max_{\substack{0\le\Delta q<5,\\ 0\le\Delta r<5}} O^{[1]}_{p(q_s+\Delta q)(r_s+\Delta r)}$$

$$\therefore \frac{\partial \mathcal{L}}{\partial O^{[1]}_{ijk}} = \sum_{pqr} \frac{\partial \mathcal{L}}{\partial O^{[2]}_{pqr}} \frac{\partial O^{[2]}_{pqr}}{\partial O^{[1]}_{ijk}} = \sum_{pqr} \frac{\partial \mathcal{L}}{\partial O^{[2]}_{pqr}} 1\left\{ \begin{array}{ll} p=i & \text{and} \\ (j,k) = \underset{\substack{q'=q+\Delta q\\ r'=r+\Delta r}}{\arg\max} & O^{[1]}_{iq'r'} \end{array} \right\}$$

< Convolution >

$$O^{[1]}_{pqr} = \sum_{\Delta h=1}^{f_h} \sum_{\Delta w=1}^{f_w} \sum_{c=1}^{C} W^{[1]}_{pc\Delta w\Delta H} X^{(i)}_{c(q+\Delta w-1)(r+\Delta h-1)} + b^{[1]}_{p}$$

$$\frac{\partial \mathcal{L}}{\partial W^{[1]}_{fcij}} = \sum_{pqr} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{pqr}} \frac{\partial O^{[1]}_{pqr}}{\partial W^{[1]}_{fcij}} = \sum_{qr} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{fqr}} \frac{\partial O^{[1]}_{fqr}}{\partial W^{[1]}_{fcij}}$$

$$= \sum_{q}^{C_w} \sum_{r}^{C_h} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{fqr}} X^{(i)}_{c(q+i-1)(r+j-1)}$$

$$\frac{\partial \mathcal{L}}{\partial X^{(i)}_{cij}} = \sum_{pqr} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{pqr}} \frac{\partial O^{[1]}_{pqr}}{\partial X^{(i)}_{cij}} = \sum_{p} \sum_{q,r} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{pqr}} \cdot W^{[1]}_{pc(i-q+1)(j-r+1)} 1\left\{ \begin{array}{l} (i-q+1) > 0 \\ (\ \ //\ \ ) \le f_w \\ (j-r+1) > 0 \\ (\ \ //\ \ ) \le f_h \end{array} \right\}$$

$$\left(\begin{array}{l} i = q+\Delta w-1 \\ \Delta w = i-q+1. \end{array} \right)$$

$$\frac{\partial \mathcal{L}}{\partial b^{[1]}_{i}} = \sum_{pqr} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{pqr}} \frac{\partial O^{[1]}_{pqr}}{\partial b^{[1]}_{i}} = \sum_{qr} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{iqr}} \cdot \frac{\partial O^{[1]}_{iqr}}{\partial b^{[1]}_{i}} = \sum_{qr} \frac{\partial \mathcal{L}}{\partial O^{[1]}_{iqr}} \cdot 1.$$

2.

<Data>

회차	상태	처방	보상(효과)
1	S_1	a_1	$R(S_1, a_1) = 80$
2	S_1	a_2	$R(S_1, a_2) = 100$
3	S_2	a_1	$R(S_2, a_1) = 60$
4	S_3	a_1	⋮
⋮	⋮	⋮	⋮

알고있는 것: $p(s)$: state 분포

$R(s,a)$: 보상 함수.

$\pi_0(s,a)$: 기존 데이터가 가지고 있는 policy (확률).

$= p(a|s, \pi_0)$

※ $\pi : S \mapsto A$ (x)

$\pi : S \times A \mapsto R \in (0,1)$

구해야 하는 것: π_1 (새로운 policy)을 따랐을 때, $E_{\substack{s\sim p(s) \\ a\sim\pi_1(s,a)}}[R(s,a)]$ 는?

(a)

$$E_{\substack{s\sim p(s) \\ a\sim\pi_0(s,a)}}\left[\frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)} R(s,a)\right] = \sum_{(s,a)} R(s,a) \cdot \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)} \cdot p(s)\pi_0(s,a)$$

if $\hat{\pi}_0 = \pi_0$, $\sum_{(s,a)} R(s,a) \frac{\pi_1(s,a)}{\pi_0(s,a)} \cdot p(s)\pi_0(s,a)$

$$= \sum_{(s,a)} R(s,a)\, p(s)\pi_1(s,a)$$

$$= E_{\substack{s\sim p(s) \\ a\sim\pi_1(s,a)}}[R(s,a)]$$

(b) $\underset{\substack{s\sim p(s)\\ a\sim\pi_0(s,a)}}{E}\left[\frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}R(s,a)\right] \Big/ \underset{\substack{s\sim p(s)\\ a\sim\pi_0(s,a)}}{E}\left[\frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}\right]$

$$= \frac{\sum_{(s,a)} \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)} R(s,a)p(s)\pi_0(s,a)}{\sum_{(s,a)} \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)} p(s)\pi_0(s,a)} \qquad \text{if } \hat{\pi}_0 = \pi_0$$

$$= \frac{\sum_{(s,a)} R(s,a)\,p(s)\pi_1(s,a)}{\sum_{(s,a)} p(s)\pi_1(s,a)} = \frac{\underset{\substack{s\sim p(s)\\ a\sim\pi_1(s,a)}}{E}[R(s,a)]}{\underset{\substack{s\sim p(s)\\ a\sim\pi_1(s,a)}}{E}[1]} = \underset{\substack{s\sim p(s)\\ a\sim\pi_1(s,a)}}{E}[R(s,a)]$$

(c) $\underset{\pi_1}{E}[R(s,a)]$ 의 추정값으로 $\hat{V} = E_{\pi_0}\left[\frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}R(s,a)\right] / E_{\pi_0}\left[\frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}\right]$ 사용.

$E[\hat{V}] = E_{\pi_1}[R(s,a)]$ 이면 unbiased,

예시, 한 개의 example data (π_0) 가 존재한다면.

$\hat{V} = R(s,a)$. $\therefore E[\hat{V}] = \underset{\substack{s\sim p(s)\\ a\sim\pi_0(s,a)}}{E}[R(s,a)]$ ($R(s,a)$는 π_0에서 샘플링 되었으므로)

$\underset{\substack{s\sim p(s)\\ a\sim\pi_0(s,a)}}{E}[R(s,a)] \neq \underset{\substack{s\sim p(s)\\ a\sim\pi_1(s,a)}}{E}[R(s,a)]$ 이므로 (일반적으로 $\pi_0 \neq \pi_1$ 이므로)

biased 되기 쉬움.!

(d)

(i) $$E_{\substack{s\sim p(s) \\ a\sim\pi_0(s,a)}}\left[E_{a\sim\pi_1(s,a)}[\hat{R}(s,a)] + \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}(R(s,a)-\hat{R}(s,a))\right]$$

$$= E_{\substack{s\sim p(s) \\ a\sim\pi_1(s,a)}}[\hat{R}(s,a)] + E_{\substack{s\sim p(s) \\ a\sim\pi_1(s,a)}}[R(s,a)] - E_{\substack{s\sim p(s) \\ a\sim\pi_1(s,a)}}[\hat{R}(s,a)]$$

$$= E_{\substack{s\sim p(s) \\ a\sim\pi_1(s,a)}}[R(s,a)]$$

(ii) $$E_{\substack{s\sim p(s) \\ a\sim\pi_0(s,a)}}\left[E_{a\sim\pi_1(s,a)}[R(s,a)] + \frac{\pi_1(s,a)}{\pi_0(s,a)}(R(s,a)-R(s,a))\right]$$

$$= E_{\substack{s\sim p(s) \\ a\sim\pi_1(s,a)}}[R(s,a)]$$

(e)

(i) important sampling estimator, easier to estimate $\hat{\pi}_0$

(ii) regression estimator, easier to estimate $\hat{R}(s,a)$

3.

$$\arg\min_u \sum_{i=1}^{m} \| x^{(i)} - f_u(x^{(i)}) \|^2, \qquad \|x^{(i)}\|^2 = \|f_u(x^{(i)})\|^2 + \|x^{(i)} - f_u(x^{(i)})\|^2$$

$$= \arg\min_u \sum_{i=1}^{m} \|x^{(i)}\|^2 - \|f_u(x^{(i)})\|^2$$

$$= \arg\max_u \sum_{i=1}^{m} \|f_u(x^{(i)})\|^2$$

$$= \arg\max_u \sum_{i=1}^{m} \| uu^T x^{(i)} u \|^2$$

$$= \arg\max_u \sum_{i=1}^{m} (u^T x^{(i)})^2 \|u\|^2 \qquad \|u\| = 1.$$

$$= \arg\max_u \sum_{i=1}^{m} (u^T x^{(i)})^2$$

4.

(a)

$$l(W) = \sum_{i=1}^{n} \left(\log|W| + \sum_{j=1}^{d} \log g'(w_j^T x^{(i)}) \right)$$

$$s_j^{(i)} = w_j^T x^{(i)} \sim \mathcal{N}(0,1). \qquad \therefore g'(s) = \frac{1}{\sqrt{2\pi}} \exp(-s_j^2/2)$$

$$l(W) = \sum_{i=1}^{n} \left(\log|W| + \sum_{j=1}^{d} \left(-\frac{(w_j^T x^{(i)})^2}{2} - \frac{1}{2}\log 2\pi \right) \right)$$

$$= n \log|W| - \frac{1}{2} \sum_{i}^{n} \sum_{j}^{d} (w_j^T x^{(i)})^2 - \frac{nd}{2} \log 2\pi$$

$$= n \log|W| - \frac{1}{2} \sum_{i}^{n} \|W x^{(i)}\|^2 + C.$$

$$\nabla_W l(W) = n \cdot (W^T)^{-1} - W \sum_{i}^{m} x^{(i)} x^{(i)T} = 0.$$

$$n (W^T)^{-1} = W \sum_{i}^{n} x^{(i)} x^{(i)T}$$

$$\therefore\ n\left(\sum_{i=1}^{n} x^{(i)}x^{(i)T}\right)^{-1} = W^TW$$

$$n\,X^TX = W^TW$$

〈회전불변성〉

임의의 orthogonal matrix R on cube $W' = RW$ 라 하면,

$$W'^TW' = (RW)^TRW = W^TR^TRW = W^TW$$

즉, 위 조건 $nX^TX = W^TW$ 를 만족하는 W 는 얼마든지 만들어낼 수 있다.

$\Rightarrow$ Solution이 하나가 아니므로 gaussian sample은 부적절.

(b)

$$l(W) = \sum_{i}^{n}\left(\log|W| + \sum_{j}^{d}\log g'(W_j^Tx^{(i)})\right)$$

$$g'(s) = \frac{1}{2}\exp(-|s|)$$

$$\therefore\ l(W) = n\log|W| - \sum_{i}^{n}\sum_{j}^{d}|W_j^Tx^{(i)}| + C$$

$$= n\log|W| - \sum_{i}^{n}\|Wx^{(i)}\|_1 + C$$

$$\nabla l(W) = n(W^{-1})^T - \sum_{i}^{n} sign(Wx^{(i)})x^{(i)T}$$

$$\nabla l(W) = (W^{-1})^T - sign(Wx^{(i)})x^{(i)T} \quad (single\ example)$$

$$\therefore\ W := W + \alpha\left((W^{-1})^T - sign(Wx^{(i)})x^{(i)T}\right)$$

5. (a)

$$\|B(V_1) - B(V_2)\|_\infty \le \gamma \|V_1 - V_2\|_\infty$$

$$\|B(V_1) - B(V_2)\|_\infty = \max_s |V_1'(s) - V_2'(s)| = \max_s \left| \gamma \max_a \sum_{s'} P_{sa}(s') V_1(s') - \gamma \max_a \sum_{s'} P_{sa}(s') V_2(s') \right|$$

$$\left(\because \left| \max_a f(a) - \max_a g(a) \right| \le \max_a |f(a) - g(a)| \right)$$

$$\le \gamma \max_s \max_a \left| \sum_{s'} P_{sa}(s') (V_1(s') - V_2(s')) \right|$$

$$\le \gamma \max_s \max_a \sum_{s'} P_{sa}(s') |V_1(s') - V_2(s')| \quad (\because |A+B| \le |A| + |B|)$$

$$\le \gamma \max_s \max_a \sum_{s'} P_{sa}(s') \max_{s'} |V_1(s') - V_2(s')|$$

$$= \gamma \underbrace{\max_s \max_a \sum_{s'} P_{sa}(s')}_{=1} \|V_1 - V_2\|_\infty = \gamma \|V_1 - V_2\|_\infty$$

motemote

(b) $\|B(V_1) - B(V_2)\|_\infty = \|V_1 - V_2\|_\infty \le \gamma \|V_1 - V_2\|_\infty$

$\because \|V_1 - V_2\|_\infty = 0.$

$\therefore V_1 = V_2$